

Question ID: 1fc4f47b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Medium

Question

At a movie theater, there are a total of **350** customers. Each customer is located in either theater A, theater B, or theater C. If one of these customers is selected at random, the probability of selecting a customer who is located in theater A is **0.48**, and the probability of selecting a customer who is located in theater B is **0.24**. How many customers are located in theater C?

Answer

- A. **28**
- B. **40**
- C. **84**
- D. **98**

Correct Answer: D

Rationale

Choice D is correct. It’s given that at a movie theater, there are a total of **350** customers and that each customer is located in either theater A, theater B, or theater C. If the probability of selecting a customer in theater A is **0.48**, then **(0.48)(350)**, or **168**, customers are located in theater A. If the probability of selecting a customer in theater B is **0.24**, then **(0.24)(350)**, or **84**, customers are located in theater B. It follows that there are **168 + 84**, or **252**, customers in theater A and theater B. Therefore, there are **350 – 252**, or **98**, customers in theater C.

Choice A is incorrect. This is the percent, not the number, of the customers that are located in theater C.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of customers that are located in theater B, not theater C.